



MANAGE PETRO
FUEL MANAGEMENT SOLUTION

Manage Petro

Project Management Dashboard Design

Prepared by:

Hadi (Harvey) Afshari | MBA, DBA | Project Manager

Version: 14

May 21, 2026

Table of Contents

<i>PMO Dashboard Structure</i>	<i>4</i>
Category Definition Rules	4
Important Rule	4
Metric Ownership Rule	4
Dashboard Design Principle	5
<i>Dashboard Metric Categories</i>	<i>6</i>
1. Delivery Health	6
2. Execution Efficiency.....	7
3. Quality & Returns.....	8
4. Bug Intelligence.....	9
5. Planning & Estimation Discipline.....	9
6. Release Readiness	10
7. Ticket Quality.....	11
<i>Cross-Metrics Analytical Layer</i>	<i>12</i>
Purpose	12
Analytical Cross-Metrics.....	12
<i>Advanced Quality Intelligence Based on Custom Fields.....</i>	<i>14</i>
Complexity Weight	14
Return Reason	14
Bug Source.....	15
<i>Phase-Based Implementation Approach</i>	<i>16</i>
Phase 1: Control Signals & Operational Visibility.....	16
Phase 2: Diagnostic & Root Cause Analysis	18
Phase 3: Advanced Analysis & Decision Support	21
Phase-Based Metric Distribution Matrix	23
<i>Phase 1 Implementation Plan</i>	<i>24</i>
Objective	24
Step 1: Data Readiness & Enforcement	24

Step 2: Workflow Validation.....	25
Step 3: Define and Build Core Metrics.....	25
Step 4: Dashboard Setup	27
Step 5: Alerts & Control Mechanisms.....	27
Step 6: Validation & Calibration.....	28
Step 7: Adoption & Training	29
Step 8: Continuous Feedback Loop	29
Phase 1 Selected Metrics.....	30
<i>Phase 1 Dashboard Design for Implementation.....</i>	<i>32</i>
1. Design Approach	32
2. Dashboard Structure & Rules.....	33
2.1 Phase 1 Dashboard Pages	33
2.2 Target Dashboard Tables Hierarchy	34
2.3 Metric and Table Placement	37
2.4 Global Filters	39
2.5 Report Modes	40
2.6 Drill-Down Rules.....	40
2.7 Color and Risk Logic.....	41
3. Dashboard Details	42
3.1 PMO Control Dashboard	42
3.2 Sprint Delivery Dashboard	44
3.3 Team / Ticket Owner Performance Dashboard.....	49
3.4 QA & Returns Dashboard	52
3.5 Bug Control Dashboard	55
3.6 Planning & Estimation Dashboard	57
3.7 Ticket Quality & Data Discipline Dashboard.....	60
3.8 Time Utilization & Scope Control Dashboard.....	62
4. Alert Rules	64
5. Phase 1 Dashboard Tables Inventory	66
6. Glossary / Key Definitions.....	69
7. Implementation Notes	70
8. Final Expected Outcome	72

PMO Dashboard Structure

Category Definition Rules

Each dashboard category represents a distinct management perspective:

- **Delivery Health** → Output performance, delivery reliability, sprint commitment, milestone progress, and technical debt.
- **Quality & Returns** → Internal failure loops, returns, rework, return causes, and quality signals before defects become external bugs.
- **Execution Efficiency** → Time usage, flow efficiency, blockers, idle time, status duration, logged effort, and work-density signals during execution.
- **Planning & Estimation Discipline** → Accuracy and realism of planning inputs before and during execution, including estimation, commitment, not-started work, scope control, and out-of-sprint work.
- **Bug Intelligence** → Bug creation, resolution, source, leakage, severity, regression, test escape, and defect lifecycle behavior.
- **Release Readiness** → Go/no-go indicators for release decisions and readiness risk.
- **Ticket Quality** → Data completeness, update discipline, field accuracy, and reliability of dashboard source data.

Important Rule

Each metric is assigned to only one official category based on its primary management purpose, even if the metric may be reused in multiple dashboards.

Metric Ownership Rule

- **Delivery metrics** → PM / Team Lead
- **Quality metrics** → QA Lead
- **Execution Efficiency metrics** → Team Lead / Dev Lead
- **Planning metrics** → PM
- **Bug metrics** → QA Lead + Dev Lead
- **Release Readiness metrics** → PMO / PM / QA Lead

- Ticket Quality metrics → PMO

Dashboard Design Principle

Dashboard pages do not need to follow metric categories exactly. A dashboard may combine metrics from different categories when this helps a stakeholder make better decisions. However, the official metric category remains fixed in the metrics table.

Dashboard Metric Categories

1. Delivery Health

Goal: Measure whether the team is delivering planned scope on time, meeting sprint commitments, progressing toward milestones, and controlling accumulated unfinished committed work.

Metrics

- Completed Committed Tickets
 - Sprint Completion Rate (%)
 - High-Priority Delivery Rate (%)
 - Planned Tickets
 - Carry-over (%)
 - Throughput
 - Carry-over Trend
 - Delivery Predictability (%)
 - Milestone Deliverables
 - Milestone Completion (%)
 - Sprint Status
 - Technical Debt (Count)
 - Technical Debt Ratio (%)
 - Technical Debt Aging
 - New Debt Created
 - Debt Burn-down
-

2. Execution Efficiency

Goal: Measure how efficiently time is utilized, how work flows through the system, where bottlenecks exist, and how logged effort is distributed.

Metrics

- Cycle Time
- High-Priority Cycle Time
- Lead Time
- Blocked Tickets Count
- Blocked Duration
- Ticket Idle Time
- Overdue Tickets
- Overdue Duration
- Time in Status
- Integration Time
- Active Dev Time
- Avg Dev Time per Ticket
- Tickets Stuck in Status
- Work-in-Progress (WIP)
- Bottleneck Stage
- Complexity-Normalized Cycle Time
- Sprint Logged Time
- Non-Sprint Logged Time
- Work Intensity Ratio (%)

3. Quality & Returns

Goal: Measure how often work is completed correctly, identify internal failure loops, and analyze rework patterns before defects become external bugs.

Clarification

- **Returns ≠ Bugs**
- **Returns** = internal failure loops during delivery, QA, review, merge, or final check.
- **Bugs** = defect records or externalized issues captured as bug items.

Metrics

- Return Rate (%)
- Returns per Ticket
- Avg Returns per Ticket
- Return Loop Depth
- Return Cycle Duration
- Return-Induced Dev Time
- Rework Ratio (%)
- Total Rework Time
- First-Time Pass Rate (%)
- QA Fail Rate (%)
- Validation Rejection Rate (%)
- High-return Tickets
- Return Trend
- Return Reasons Distribution
- Return-based Fail Rate (%)
- High Complexity Failure Rate (%)
- Critical Return Reason Rate (%)

4. Bug Intelligence

Goal: Analyze bug trends, root causes, source categories, severity, leakage, and resolution effectiveness.

Clarification

- **Returns ≠ Bugs**
- Bugs are tracked as defect records and should not be mixed with internal return-loop metrics.

Metrics

- Bugs Created
- Bugs Resolved
- Bug Backlog Trend
- Bug Leakage (%)
- Bug Reopen Rate
- Severity Distribution
- Critical Bugs Open
- Open Critical Bug Aging
- Resolution Time by Severity
- QA Escape Rate (%)
- Test Coverage Gap Rate (%)
- Regression Rate (%)

5. Planning & Estimation Discipline

Goal: Evaluate planning accuracy, estimation reliability, commitment realism, out-of-sprint work, carry-over causes, and execution alignment with planning assumptions.

Metrics

- Planned vs Actual Time
- Estimation Accuracy (%)
- Planned vs Actual Effort
- Overrun Ratio
- Underestimation Frequency
- Commitment Reliability (%)
- Scope Change Rate
- Scope Volatility Trend
- Re-prioritized Tickets
- Repeated Carry-over
- Not Started (Count & %)
- Requirement Quality Issue Rate (%)
- Out-of-Sprint Work Ratio (%)
- PM vs Owner Estimation Variance (%)
- Owner Estimation Accuracy (%)
- Carry-over Reason Distribution

Note on Commitment Reliability

Commitment Reliability (%) is intentionally similar to Sprint Completion Rate (%), but it is interpreted from a planning discipline perspective. Sprint Completion Rate belongs to Delivery Health, while Commitment Reliability belongs to Planning & Estimation Discipline.

6. Release Readiness

Goal: Assess whether the system is ready for release with acceptable risk.

Metrics

- Release Readiness Score
 - Testing Coverage (%)
-

7. Ticket Quality

Goal: Ensure ticket data is complete, accurate, and reliable enough for planning, execution control, performance analysis, and dashboard interpretation.

Metrics

- Missing Required Fields (%)
 - Tickets Missing Estimation (%)
 - Tickets Missing Owner (%)
 - Tickets Missing Priority (%)
 - Tickets Without Activity Logs (%)
 - Tickets Without Updates (%)
 - Time Logging Completeness (%)
 - Description Quality Rate (%)
 - Status Inconsistency Rate
-

Cross-Metrics Analytical Layer

Purpose

Cross-metrics combine multiple primary metrics to provide root-cause insights and management-level interpretation beyond surface-level reporting.

Cross-metrics are not standalone KPI definitions in the same way as base metrics. They are analytical relationships used for:

- diagnosing issues
- identifying patterns
- supporting management decisions
- balancing interpretation across delivery, quality, planning, execution, and capacity signals

Analytical Cross-Metrics

Cross-Metric	Description
Cycle Time vs Complexity Weight	Determines whether higher complexity is properly reflected in delivery time.
Rework Ratio by Developer	Identifies developers or owners producing unstable outputs or requiring coaching/support.
Returns Distribution by Module / Feature	Detects unstable product areas and recurring quality weaknesses.
Bug Leakage vs Return Rate	Distinguishes whether quality problems are being caught internally or escaping as bugs.
Bottleneck Stage Analysis	Identifies the workflow stage where work slows down the most.
Estimation Accuracy vs Carry-over	Determines whether planning inaccuracies contribute to unfinished commitments.
WIP vs Cycle Time	Evaluates whether too much active work is slowing delivery.
Cycle Time vs Return Rate	Checks whether faster delivery is causing lower quality or more returns.
Flow Efficiency (%)	Measures productive work time compared with total lifecycle time.
QA Effectiveness Index	Evaluates whether QA catches issues early or allows escapes.

Cross-Metric	Description
Release Risk Score	Combines release blockers, quality, bug, and coverage indicators for release decision support.
Workload vs Rework	Detects whether overloaded owners or teams produce more rework.
Delivery vs Capacity	Detects utilization and delivery balance against available capacity.
High-Priority Responsiveness	Measures responsiveness to urgent or critical work.
Technical Debt Pressure	Evaluates whether unfinished committed work is accumulating or being reduced.
Planning Failure Signal	Combines not-started work and carry-over to identify planning weakness.
QA vs Dev Responsibility Balance	Separates QA escape issues from development quality issues.
Complexity vs Quality	Detects whether complex tickets drive returns, bugs, or leakage.
Regression Risk Indicator	Detects instability caused by regressions and repeated rework.

Advanced Quality Intelligence Based on Custom Fields

The dashboard uses selected custom fields to improve root-cause analysis and performance interpretation.

Complexity Weight

Complexity Weight is a custom field that stores a Fibonacci-based complexity value. Allowed values:

- 5
- 8
- 13
- 21

It is used to:

- normalize cycle time
- compare delivery performance fairly across different ticket complexities
- detect tickets with low complexity but unusually long cycle time
- identify whether complex tickets are more likely to cause returns, bugs, or leakage

Return Reason

Return Reason is used to classify why a ticket is returned.

Allowed reasons:

- Wrong Implementation
- Incomplete Fix
- Caused Side Effect
- Missed Requirement / Misunderstanding
- Data / Environment Issue
- Code Quality Issue
- Merge / Integration Conflict
- Out-of-Scope Modification

It is used to:

- identify dominant causes of internal failure
- support QA and developer improvement actions
- distinguish development, requirement, environment, merge, and scope-related return causes

- calculate Critical Return Reason Rate (%)

Bug Source

Bug Source identifies the likely origin of a bug.

Allowed sources:

- Design / Requirement Gap
- Side Effect / Regression
- Test Escape
- Test Coverage Gap
- Data / Environment Issue

It is used to:

- separate QA escape problems from test coverage problems
- identify regressions and side effects
- identify requirement or design gaps
- support QA, product, and development process improvement

These fields strengthen Phase 2 diagnostic capability and support more accurate performance improvement actions.

Phase-Based Implementation Approach

Phase 1: Control Signals & Operational Visibility

Objective

Provide immediate visibility into execution, quality, planning discipline, ticket quality, and system health.

Theme

“What is happening now?”

Characteristics

- Focuses on real-time and near real-time indicators.
- Includes essential KPIs for daily and sprint-level monitoring.
- Requires minimal calculation complexity.
- Uses reliable and consistently available data.
- Provides the operational control layer before deeper diagnostic analysis.

Phase 1 Core Metrics

Delivery:

- Completed Committed Tickets
- Planned Tickets
- Carry-over (%)
- Throughput
- Sprint Completion Rate (%)
- High-Priority Delivery Rate (%)
- Technical Debt (Count)

Quality:

- Return Rate (%)

- Rework Ratio (%)
- First-Time Pass Rate (%)

Execution:

- Cycle Time
- High-Priority Cycle Time
- Blocked Tickets Count
- Ticket Idle Time
- Sprint Logged Time
- Non-Sprint Logged Time

Planning:

- Estimation Accuracy (%)
- Commitment Reliability (%)
- Not Started (Count & %)
- Out-of-Sprint Work Ratio (%)

Bugs:

- Bugs Created
- Bugs Resolved
- Bug Leakage (%)
- Critical Bugs Open

Ticket Quality:

- Missing Required Fields (%)
- Tickets Without Updates (%)

Outcome

Phase 1 enables immediate detection of:

- delivery delays
- carry-over and technical debt
- quality issues
- execution bottlenecks
- out-of-sprint effort leakage
- data gaps
- high-priority delivery failures

It establishes a strong control layer and forms the foundation for diagnostic and analytical capabilities.

Phase 2: Diagnostic & Root Cause Analysis

Objective

Enable deeper analysis and identification of root causes.

Theme

“Why is this happening?”

Characteristics

- Introduces segmented and analytical metrics.
- Requires consistent data capture and classification.
- Focuses on patterns, bottlenecks, inefficiencies, and root causes.
- Uses custom fields, time logs, workflow history, and ticket classification data.

Phase 2 Coverage Across Categories

Delivery Health:

- Carry-over Trend
- Delivery Predictability (%)
- Technical Debt Ratio (%)

- New Debt Created
- Debt Burn-down

Quality & Returns:

- Return reasons
- Return distribution
- QA Fail Rate (%)
- Validation Rejection Rate (%)
- Return Trend
- High Complexity Failure Rate (%)
- Critical Return Reason Rate (%)

Execution Efficiency:

- Lead Time
- Overdue Tickets
- Overdue Duration
- Time in Status
- Integration Time
- Active Dev Time
- Complexity-Normalized Cycle Time
- Work Intensity Ratio (%)

Planning & Estimation Discipline:

- Scope Change Rate
- Overrun Ratio
- Underestimation Frequency
- Requirement Quality Issue Rate (%)

- PM vs Owner Estimation Variance (%)
- Owner Estimation Accuracy (%)
- Carry-over Reason Distribution

Bug Intelligence:

- Bug Backlog Trend
- Severity Distribution
- Bug Reopen Rate
- QA Escape Rate (%)
- Test Coverage Gap Rate (%)
- Regression Rate (%)

Release Readiness:

- Testing Coverage (%)

Ticket Quality:

- Time Logging Completeness (%)
- Description Quality Rate (%)

Outcome

Phase 2 enables identification of:

- root causes of delivery delays
- sources of quality degradation
- workflow inefficiencies
- estimation gaps
- QA, development, requirement, and regression issues
- out-of-sprint workload pressure
- technical debt accumulation causes

Phase 3: Advanced Analysis & Decision Support

Objective

Support optimization, forecasting, and strategic decision-making.

Theme

“What should we improve or decide?”

Characteristics

- Includes advanced, aggregated, and trend-based metrics.
- Enables cross-metric analysis and performance evaluation.
- Supports management-level and release-level decisions.

Phase 3 Coverage Across Categories

Delivery Health:

- Milestone Completion (%)
- Technical Debt Aging

Quality & Returns:

- Return Loop Depth
- Return Cycle Duration
- High-return Tickets
- Return-based Fail Rate (%)

Execution Efficiency:

- Work-in-Progress (WIP)
- Bottleneck Stage

Planning & Estimation Discipline:

- Scope Volatility Trend

- Re-prioritized Tickets
- Repeated Carry-over

Bug Intelligence:

- Open Critical Bug Aging
- Resolution Time by Severity

Release Readiness:

- Release Readiness Score

Ticket Quality:

- Status Inconsistency Rate

Outcome

Phase 3 improves:

- delivery predictability
 - planning maturity
 - release readiness decisions
 - system stability before release
 - long-term process optimization
-

Phase-Based Metric Distribution Matrix

Category	Phase 1	Phase 2	Phase 3
Delivery Health	Completed Committed Tickets; Sprint Completion Rate (%); Planned Tickets; Carry-over (%); Throughput; High-Priority Delivery Rate (%); Technical Debt (Count); Sprint Status	Carry-over Trend; Delivery Predictability (%); Milestone Deliverables; Technical Debt Ratio (%); New Debt Created; Debt Burn-down	Milestone Completion (%); Technical Debt Aging
Execution Efficiency	Cycle Time; High-Priority Cycle Time; Blocked Tickets Count; Blocked Duration; Ticket Idle Time; Sprint Logged Time; Non-Sprint Logged Time	Lead Time; Overdue Tickets; Overdue Duration; Time in Status; Integration Time; Active Dev Time; Avg Dev Time per Ticket; Tickets Stuck in Status; Complexity-Normalized Cycle Time; Work Intensity Ratio (%)	Work-in-Progress (WIP); Bottleneck Stage
Quality & Returns	Return Rate (%); Rework Ratio (%); Total Rework Time; First-Time Pass Rate (%)	Returns per Ticket; Avg Returns per Ticket; Return-Induced Dev Time; QA Fail Rate (%); Validation Rejection Rate (%); Return Trend; Return Reasons Distribution; High Complexity Failure Rate (%); Critical Return Reason Rate (%)	Return Loop Depth; Return Cycle Duration; High-return Tickets; Return-based Fail Rate (%)
Bug Intelligence	Bugs Created; Bugs Resolved; Bug Leakage (%); Critical Bugs Open	Bug Backlog Trend; Bug Reopen Rate; Severity Distribution; QA Escape Rate (%); Test Coverage Gap Rate (%); Regression Rate (%)	Open Critical Bug Aging; Resolution Time by Severity
Planning & Estimation Discipline	Planned vs Actual Time; Estimation Accuracy (%); Commitment Reliability (%); Not Started (Count & %); Out-of-Sprint Work Ratio (%)	Planned vs Actual Effort; Overrun Ratio; Underestimation Frequency; Scope Change Rate; Requirement Quality Issue Rate (%); PM vs Owner Estimation Variance (%); Owner Estimation Accuracy (%); Carry-over Reason Distribution	Scope Volatility Trend; Re-prioritized Tickets; Repeated Carry-over
Release Readiness	—	Testing Coverage (%)	Release Readiness Score
Ticket Quality	Missing Required Fields (%); Tickets Without Updates (%)	Time Logging Completeness (%); Description Quality Rate (%)	Status Inconsistency Rate

Phase 1 Implementation Plan

Objective

Establish a reliable operational control layer that provides near real-time visibility into execution, delivery, quality, planning discipline, ticket quality, and time utilization.

Step 1: Data Readiness & Enforcement

Ensure all required data is consistently captured before building dashboards.

Mandatory Fields

- Ticket Owner
- Sprint
- Original Sprint
- Estimated Time
- PM Estimation
- Owner Estimation
- Start Date
- Due Date
- Priority
- Complexity Weight
- Carry-over Reason, where applicable
- Return Reason, where applicable
- Bug Source, where applicable

Workflow Discipline

- No work should happen outside defined statuses.
- Status changes must be made promptly and accurately.

- Return loops must be captured consistently.
- Pending and On-hold must be used consistently.
- Done and Archived must clearly indicate completion states.

Time Logging Discipline

- Team members must log time regularly.
 - Time logs must be linked to tickets.
 - Time logs must include activity type where applicable.
 - Time logs must allow separation between sprint work and non-sprint work.
-

Step 2: Workflow Validation

Validate that Ticket Journey Mapping is implemented correctly.

Validation should confirm:

- all status transitions are controlled
 - all return loops are tracked
 - timestamps are reliably recorded
 - cycle time can be calculated
 - lead time can be calculated
 - return count can be calculated
 - time in status can be calculated
 - no workflow stages are bypassed without visibility
-

Step 3: Define and Build Core Metrics

Implement Phase 1 control metrics first.

Key Metric Groups

Delivery:

- Completed Committed Tickets
- Planned Tickets
- Carry-over (%)
- Throughput
- Sprint Completion Rate (%)
- High-Priority Delivery Rate (%)
- Technical Debt (Count)

Quality:

- Return Rate (%)
- Rework Ratio (%)
- First-Time Pass Rate (%)

Efficiency:

- Cycle Time
- High-Priority Cycle Time
- Blocked Tickets Count
- Ticket Idle Time
- Sprint Logged Time
- Non-Sprint Logged Time

Planning:

- Estimation Accuracy (%)
- Commitment Reliability (%)
- Not Started (Count & %)
- Out-of-Sprint Work Ratio (%)

Bugs:

- Bugs Created
- Bugs Resolved
- Bug Leakage (%)
- Critical Bugs Open

Ticket Quality:

- Missing Required Fields (%)
 - Tickets Without Updates (%)
-

Step 4: Dashboard Setup

Create role-based dashboards:

- PMO Control Dashboard
 - Sprint Delivery Dashboard
 - Team / Ticket Owner Performance Dashboard
 - QA & Returns Dashboard
 - Bug Control Dashboard
 - Planning & Estimation Dashboard
 - Ticket Quality & Data Discipline Dashboard
 - Time Utilization & Scope Control Dashboard
-

Step 5: Alerts & Control Mechanisms

Configure alerts for critical conditions:

- low sprint completion
- high carry-over

- technical debt growth
 - blocked work
 - high return rate
 - high rework time
 - critical bugs open
 - bug growth greater than bug closure
 - not-started committed work
 - high out-of-sprint work ratio
 - missing required fields
 - tickets without updates
 - high-priority delivery delay
-

Step 6: Validation & Calibration

Run dashboards in parallel with actual sprint work for 2–3 weeks.

Validate:

- formulas
 - filters
 - data quality
 - status usage
 - time-log usage
 - sprint relation and original sprint logic
 - carry-over reason usage
 - Redmine extraction logic
-

Step 7: Adoption & Training

Train PMs, team leads, QA, and ticket owners on:

- how to update tickets
- how to log time
- how to use status transitions
- how to select return reasons
- how to select bug sources
- how to classify carry-over reason
- how metrics should be interpreted

Clarify that metrics are intended for improvement, coaching, planning, and transparency — not automatic punishment.

Step 8: Continuous Feedback Loop

Collect feedback from:

- PM
- PMO
- team leads
- QA
- developers
- ticket owners

Use feedback to improve:

- dashboard usability
 - formulas
 - filters
 - field definitions
 - thresholds
 - Phase 2 diagnostic readiness
-

Phase 1 Selected Metrics

#	Category	Metric	Definition & Explanation	Formula
1	Delivery Health	Completed Committed Tickets	Committed sprint tickets completed within sprint	COUNT(committed AND closed within sprint)
2	Delivery Health	Sprint Completion Rate (%)	% of committed tickets delivered	Completed Committed / Planned * 100
3	Delivery Health	Planned Tickets	Total committed sprint tickets	COUNT(committed tickets)
4	Delivery Health	Carry-over (%)	% of committed tickets not delivered	COUNT(Original Sprint = Sprint X AND Status != Done/Closed at Sprint End) / Planned Tickets * 100
6	Delivery Health	Throughput	Total completed tickets	COUNT(closed tickets)
10	Delivery Health	Sprint Status	Sprint health indicator	IF Completion $\geq$ 90% AND Carry-over < 10% THEN On Track
11	Execution Efficiency	Cycle Time	Time from In Progress to Done	AVG(close - first In Progress)
13	Execution Efficiency	Blocked Tickets Count	Blocked tickets count	COUNT(status = Blocked)
14	Execution Efficiency	Blocked Duration	Time tickets remain blocked	AVG(blocked duration)
15	Execution Efficiency	Ticket Idle Time	Time since last activity	NOW - last update
25	Quality & Returns	Return Rate (%)	Tickets with returns	Returned / total * 100
31	Quality & Returns	Rework Ratio (%)	Rework share	Rework / total dev * 100
32	Quality & Returns	Total Rework Time	Rework hours	SUM(rework time)
33	Quality & Returns	First-Time Pass Rate (%)	No-return tickets	No-return / total * 100
40	Bug Intelligence	Bugs Created	New bugs	COUNT(created bugs)
41	Bug Intelligence	Bugs Resolved	Closed bugs	COUNT(closed bugs)
43	Bug Intelligence	Bug Leakage (%)	Escaped bugs	Escaped / total * 100
46	Bug Intelligence	Critical Bugs Open	Open critical bugs	COUNT(critical open)
49	Planning & Estimation Discipline	Planned vs Actual Time	Estimate vs actual	Estimate vs actual

#	Category	Metric	Definition & Explanation	Formula
50	Planning & Estimation Discipline	Estimation Accuracy (%)	Accuracy KPI	$1 - \text{ABS}(\text{Actual} - \text{Estimate}) / \text{Estimate}$
54	Planning & Estimation Discipline	Commitment Reliability (%)	Delivery reliability	Delivered / Committed
61	Ticket Quality	Missing Required Fields (%)	Missing fields	Missing / total * 100
62	Ticket Quality	Tickets Without Updates (%)	Inactive tickets	No update / active * 100
66	Delivery Health	High-Priority Delivery Rate (%)	High-priority tickets delivered on time	High-priority completed on time / total high-priority tickets * 100
67	Execution Efficiency	High-Priority Cycle Time	Average cycle time for high-priority tickets	AVG(close date - first In Progress date for high-priority tickets)
68	Planning & Estimation Discipline	Not Started (Count & %)	Committed tickets never started during sprint	COUNT(Original Sprint = Sprint X AND never transitioned to In Progress); Count / Planned Tickets * 100
69	Delivery Health	Technical Debt (Count)	Unfinished committed tickets from previous sprints	COUNT(Original Sprint < Current Sprint AND Status != Done/Closed)
81	Execution Efficiency	Sprint Logged Time	Time logged on sprint tickets during selected period	SUM(time logs on tickets assigned to sprint)
82	Execution Efficiency	Non-Sprint Logged Time	Time logged on tickets outside sprint scope during selected period	SUM(time logs on tickets not assigned to sprint)
83	Planning & Estimation Discipline	Out-of-Sprint Work Ratio (%)	Share of logged time spent outside sprint scope	Non-Sprint Logged Time / Total Logged Time * 100

Phase 1 Dashboard Design for Implementation

1. Design Approach

The Phase 1 dashboard translates approved Phase 1 metrics into practical dashboard views for daily control, sprint monitoring, quality follow-up, bug control, planning review, ticket-owner visibility, time utilization, and data discipline.

The dashboard structure is not required to follow metric categories exactly. Metrics remain officially classified under Delivery Health, Execution Efficiency, Quality & Returns, Bug Intelligence, Planning & Estimation Discipline, Release Readiness, and Ticket Quality. However, dashboard pages are organized based on stakeholder needs and decision usefulness.

The dashboard should provide:

- high-level management insight through KPI cards, health indicators, summary charts, and risk tables
- operational drill-down through detailed tables showing tickets, owners, statuses, returns, bugs, planning gaps, logged time, and data-quality issues

Dashboard Design Principles

1. Control-first design

The dashboard should quickly answer what is happening now, where risk is building, and what needs action.

2. Stakeholder-friendly structure

Pages should be easy for PM, PMO/management, team leads, QA, and ticket owners to use.

3. Summary first, detail second

Each dashboard page should start with important signals, then allow drill-down into detailed tables.

4. Metric reuse is allowed

A metric can appear in more than one dashboard page when it supports different decisions.

5. Returns and bugs must remain separated

Returns represent internal failure or rework loops. Bugs represent defect records or externalized quality issues.

6. Performance views must be balanced

Ticket-owner performance should not be judged only by completed ticket count. It should also consider commitment, debt, cycle time, returns, priority responsiveness, complexity, time allocation, and data quality.

7. Data quality affects confidence

Missing owners, missing estimations, missing dates, missing updates, missing time logs, or unreliable status usage reduce confidence in dashboard interpretation.

2. Dashboard Structure & Rules

2.1 Phase 1 Dashboard Pages

The Phase 1 dashboard should include eight pages/tabs.

#	Dashboard Page	Main Purpose
1	PMO Control Dashboard	Overall project/sprint health, key risks, and cross-project visibility.
2	Sprint Delivery Dashboard	Sprint commitment, completion, carry-over, status distribution, and execution health.
3	Team / Ticket Owner Performance Dashboard	Owner-level workload, commitment, delivery, throughput, cycle time, returns, debt, and high-priority responsiveness.
4	QA & Returns Dashboard	Return behavior, rework, first-time pass, return reasons, and internal quality signals.
5	Bug Control Dashboard	Bug creation, closure, remaining bugs, leakage/source categories, impact, and affected pages.
6	Planning & Estimation Dashboard	Planned vs actual time, estimation deviation, commitment reliability, not-started work, carry-over reasons, and planning gaps.
7	Ticket Quality & Data Discipline Dashboard	Missing required fields, missing updates, planning-field gaps, and data reliability.
8	Time Utilization & Scope Control Dashboard	Logged effort, sprint vs non-sprint work, time allocation, work-density, and out-of-sprint scope visibility.

2.2 Target Dashboard Tables Hierarchy

The following hierarchy defines how all designed Phase 1 dashboard tables are classified under the eight Target Dashboard sections. This hierarchy is used as the reference structure for implementation, validation, and future dashboard expansion.

The table hierarchy is organized by management purpose and decision usefulness. Each dashboard section groups the tables that best support its operational or management objective. Some tables may also be reused as supporting or cross-view tables in other dashboard sections when they provide useful context.

1. PMO Control Dashboard

T01 — Open Technical Tickets Snapshot
T02 — Cross-Project Risk Summary
T03 — Open Tasks / Business Jobs
T04 — Open Milestones
T44 — Feature / Improvement Visibility by Version or Date

2. Sprint Delivery Dashboard

T05 — Committed Tickets for Sprint #X
T06 — Sprint Tasks Distribution in Statuses
T07 — Sprint Execution Efficiency / Original Sprint Composition
T08 — Sprint Completion Summary
T09 — Done Tickets Time Efficiency
T10 — Ticket Status Change Trend
T11 — Tracker Change Trend
T12 — High-Priority Delivery View
T45 — Daily Agile Board Snapshot

3. Team / Ticket Owner Performance Dashboard

T13 — Ticket Delivery Analysis by Owner
T14 — Done Committed Tickets Analysis by Owner
T15 — Average Time for Each Status
T16 — Owner Ticket Idle / No Update Detail
T17 — Owner Time Allocation

Note: T17 is a secondary/cross-view. The primary source table is under Time Utilization & Scope Control as T42.

4. QA & Returns Dashboard

- T18 — Done Tickets Analysis
- T19 — Returned Tickets Analysis — Count
- T20 — Returned Tickets Analysis — Duration
- T21 — Work & Rework Ratio
- T22 — Return Reason Analysis

5. Bug Control Dashboard

- T23 — Bugs Analysis by Category
- T24 — Bug Impact Summary
- T25 — Related Page Histogram
- T26 — Critical Bugs Open Detail

6. Planning & Estimation Dashboard

- T27 — Planning vs Actual Time Analysis
- T28 — PM Estimation vs Owner Estimation vs Time Spent
- T29 — Original Sprint Composition
- T30 — Open Planned Tickets Health
- T31 — Not Started Tickets
- T32 — Commitment Reliability Summary
- T33 — Carry-over Reason Analysis

7. Ticket Quality & Data Discipline Dashboard

- T34 — Open Planned Tickets Health
- T35 — Tickets Without Updates
- T36 — Missing Required Fields Detail
- T37 — Data Reliability by Project

8. Time Utilization & Scope Control Dashboard

- T38 — Spent Time: General vs Sprint Tickets
- T39 — Out-of-Sprint Scope Works
- T40 — Spent Time on Group of Tickets in Specific Date Period
- T41 — Average Spent Time vs Calendar Duration
- T42 — Owner Time Allocation
- T43 — Activity Distribution in Spent Time Report

Rationale

This hierarchy is rational because:

- Each dashboard section has a clear management purpose.
- Tables are grouped by the question they help answer, not only by metric category.
- Cross-view usage is allowed where the same table supports more than one management perspective.
- The hierarchy provides a simple checklist for validating dashboard coverage.
- The detailed design of each table remains in Section 3 and the full inventory remains in Section 5.

2.3 Metric and Table Placement

Each approved metric keeps its official category, but dashboard placement is based on decision usefulness.

Metric / Table	Main Dashboard	Secondary Dashboard
Completed Committed Tickets	Sprint Delivery	PMO Control; Owner Performance
Sprint Completion Rate (%)	Sprint Delivery	PMO Control; Planning & Estimation
Planned Tickets	Sprint Delivery	Planning & Estimation
Carry-over (%)	Sprint Delivery	PMO Control; Planning & Estimation
Throughput	Owner Performance	Sprint Delivery
Sprint Status	PMO Control	Sprint Delivery
High-Priority Delivery Rate (%)	PMO Control	Owner Performance
Technical Debt Count	PMO Control	Sprint Delivery; Planning & Estimation
Cycle Time	Owner Performance	Sprint Delivery
Lead Time	Sprint Delivery	Owner Performance
Blocked Tickets Count	PMO Control	Sprint Delivery
Blocked Duration	Sprint Delivery	Owner Performance
Ticket Idle Time	Owner Performance	Ticket Quality
High-Priority Cycle Time	Owner Performance	PMO Control
Return Rate (%)	QA & Returns	PMO Control; Owner Performance
Rework Ratio (%)	QA & Returns	Owner Performance
Total Rework Time	QA & Returns	Sprint Delivery
First-Time Pass Rate (%)	QA & Returns	Owner Performance
Bugs Created	Bug Control	PMO Control
Bugs Resolved	Bug Control	PMO Control
Bug Leakage (%)	Bug Control	QA & Returns
Critical Bugs Open	PMO Control	Bug Control
Planned vs Actual Time	Planning & Estimation	Sprint Delivery
Estimation Accuracy (%)	Planning & Estimation	Owner Performance
Commitment Reliability (%)	Planning & Estimation	Sprint Delivery

Metric / Table	Main Dashboard	Secondary Dashboard
Not Started Count & %	Planning & Estimation	Sprint Delivery
Missing Required Fields (%)	Ticket Quality	PMO Control
Tickets Without Updates (%)	Ticket Quality	Owner Performance
Open Technical Tickets	PMO Control	Sprint Delivery
Open Planned Tickets Health	Ticket Quality	Planning & Estimation
Open Tasks / Business Jobs	PMO Control	Planning & Estimation
Open Milestones	PMO Control	Sprint Delivery
Ticket Status Change Trend	Sprint Delivery	PMO Control
Tracker Change Trend	Sprint Delivery	Bug Control
Return Reason Analysis	QA & Returns	Owner Performance
Sprint Tasks Distribution in Statuses	Sprint Delivery	Owner Performance
Bugs Analysis by Category	Bug Control	PMO Control
Bug Impact Rating	Bug Control	PMO Control
Related Page Histogram	Bug Control	QA & Returns
Average Time for Each Status	Owner Performance	Sprint Delivery
Committed Tickets for Sprint #X	Sprint Delivery	Planning & Estimation
Spent Time: General vs Sprint Tickets	Time Utilization & Scope Control	PMO Control
PM Estimation vs Owner Estimation vs Time Spent	Planning & Estimation	Owner Performance
Out-of-Sprint Scope Works	Time Utilization & Scope Control	Planning & Estimation
Carry-over Reason Analysis	Planning & Estimation	Sprint Delivery
Average Spent Time vs Calendar Duration	Time Utilization & Scope Control	Owner Performance
Spent Time on Group of Tickets in Date Period	Time Utilization & Scope Control	Owner Performance

2.4 Global Filters

The dashboard should support these filters where applicable:

- Project
 - Sprint
 - Original Sprint
 - Current Sprint
 - Date range
 - Report date
 - Team
 - Team Lead
 - Ticket Owner
 - Assignee
 - Tracker
 - Priority
 - Status
 - Return Reason
 - Return Type
 - Bug Source / Bug Category
 - Bug Related Page / Module
 - Time Log User
 - Time Log Date
 - Time Log Activity
 - Sprint Relation
 - PM Estimation
 - Owner Estimation
 - Carry-over Reason
-

2.5 Report Modes

Report Mode	Purpose	Example Tables
State at Specific Date	Show snapshot at a point in time	Open Technical Tickets; Open Planned Tickets Health; Sprint Tasks Distribution
Periodic Report	Show activity during a selected period	Owner Performance; Bug Analysis; Planning vs Actual; Time Logged During Period
Trend / Flow Report	Show movement and change	Ticket Status Change Trend; Tracker Change Trend; Bug Change Trend

2.6 Drill-Down Rules

Each KPI or visual should allow drill-down to the related ticket list or detail table.

Dashboard Item	Drill-Down Target
Sprint Completion Rate	Committed ticket list with status, owner, due date, and completion state
Carry-over %	Unfinished committed tickets
Technical Debt Count	Previous-sprint unfinished tickets
Blocked Tickets Count	Blocked / Pending / On-hold ticket list
Return Rate	Tickets with at least one return
Rework Time	Tickets with rework duration and return source
Critical Bugs Open	Open critical bug list
Bugs Created / Resolved	Bug list by creation or closure period
Missing Required Fields	Tickets missing owner, dates, priority, estimation, or sprint
Tickets Without Updates	Active tickets with no update during threshold period
Owner Completion Rate	Owner committed and completed ticket list
High-Priority Cycle Time	High-priority ticket list with cycle time
Related Page Histogram	Bugs grouped by affected page/module
Out-of-Sprint Work Ratio	Time logs on tickets outside sprint scope
Sprint Logged Time	Time log entries on sprint-related tickets
PM vs Owner Estimation Variance	Tickets where PM and Owner estimates differ materially
Work Intensity Ratio	Tickets with calendar duration, spent time, and owner
Carry-over Reason Distribution	Carry-over ticket list grouped by reason

2.7 Color and Risk Logic

Color	Meaning	Use
Green	Normal / acceptable	No immediate action required
Yellow	Watch	Follow-up or monitoring required
Orange	Risk	PM or team lead intervention required
Red	Critical	Escalation, replanning, or urgent action required

Suggested Risk Areas

Area	Risk Signal
Delivery	Low completion rate, high carry-over, large unfinished committed scope
Execution	Blocked tickets, long blocked duration, high idle time
Quality	High return rate, low first-time pass rate, high rework time
Bugs	Critical bugs open, bugs created greater than bugs resolved, high leakage
Planning	High planned-vs-actual deviation, high not-started count, weak commitment reliability
Time Utilization	High out-of-sprint work ratio, high non-sprint logged time, low work intensity
Ticket Quality	Missing required fields, missing updates, missing owner, missing estimation

3. Dashboard Details

3.1 PMO Control Dashboard

Purpose

Provide a high-level control view for PM and management to understand whether projects, sprints, and active work are under control.

Primary Users

- Project Manager
- PMO
- Management
- Team Leads

Main Questions

- Is current execution healthy?
- Which project or sprint has the highest risk?
- Are there delivery, quality, bug, planning, time-utilization, or data-discipline issues requiring action?
- Where should PM attention go first?

Main KPI Cards

KPI	Purpose
Sprint Status	Overall sprint health indicator
Sprint Completion Rate (%)	Delivery progress against commitment
Carry-over (%)	Undelivered committed work
Technical Debt Count	Previous-sprint unfinished work
Blocked Tickets Count	Current execution blockage
Critical Bugs Open	Immediate product/release risk
Return Rate (%)	Internal quality risk
Missing Required Fields (%)	Data reliability risk
Tickets Without Updates (%)	Activity discipline risk
High-Priority Delivery Rate (%)	Responsiveness to urgent work
Out-of-Sprint Work Ratio (%)	Effort leakage outside sprint scope

Main Tables

A. Open Technical Tickets Snapshot

Date	Project	Total Open Technical Tickets	Backlog	Backlog %	Under Work	Under Work %	Stopped	Stopped %	Ongoing	Ongoing %

Technical tickets include:

- Bug
- Feature
- Change Request / Improvement

B. Cross-Project Risk Summary

Project	Sprint Status	Completion %	Carry-over %	Technical Debt	Blocked Tickets	Critical Bugs	Return Rate %	Missing Fields %	Tickets Without Updates %	Overall Risk

C. Open Tasks / Business Jobs

Project	Total Open Tasks / Business Jobs	New	To-Do	In Progress	Pending / On-Hold	Done	Risk

D. Open Milestones

Project	Milestone	Total Tickets	Completed	Pending	Returned	Completion %	Risk

E. Feature / Improvement Visibility by Version or Date

Version / Date	Project	Tracker	Feature / Improvement	Status	Owner	Sprint	Target Date	Completion State

Recommended Visuals

- KPI cards for top control signals
- Cross-project risk table
- Bar chart: total open technical tickets by project
- Stacked bar: backlog / under work / stopped / ongoing
- Risk list: critical bugs, blocked tickets, high carry-over, missing data, high out-of-sprint work

Action Triggers

- Red sprint status
- Critical bugs open
- High blocked-ticket count
- High carry-over
- Technical debt growth
- High return rate
- Missing required fields above threshold
- Tickets without updates above threshold
- High-priority tickets not completed on time
- High out-of-sprint work ratio

3.2 Sprint Delivery Dashboard

Purpose

Track sprint execution against committed scope and show where sprint work is concentrated.

Primary Users

- Project Manager
- Team Lead
- PMO

Main Questions

- Are committed sprint tickets being completed?
- What work is WIP, stopped, returned, or done?

- How much of sprint work is current vs carried-over?
- Which owners have unfinished, returned, or stopped tickets?
- Which committed tickets form the baseline for sprint metrics?

Main KPI Cards

KPI	Purpose
Planned Tickets	Total committed sprint scope
Completed Committed Tickets	Committed work completed
Sprint Completion Rate (%)	Completion against commitment
Carry-over (%)	Undelivered committed work
Throughput	Total completed tickets
Technical Debt Count	Old unfinished work
Blocked Tickets Count	Execution blockage
Blocked Duration	Time spent in blocked/stopped state
Not Started Count & %	Committed work not started
Cycle Time	Execution speed
Lead Time	End-to-end delivery time

Main Tables

A. Committed Tickets for Sprint #X

Ticket	Project	Tracker	Priority	Owner	Original Sprint	Current Sprint	Start Date	Due Date	Status	Estimate	Spent Time	Completion State	Carry-over Reason

Purpose: This is the baseline table for sprint metrics. It must be retrospectively available and should not depend only on current sprint relation after tickets are moved.

B. Sprint Tasks Distribution in Statuses

Status	Owner 1	Owner 2	Owner 3	Sum	%

Status rows:

- New
- To-Do
- In Progress
- Feedback

- Review
- Ready to Merge
- Final Check
- Returned
- Pending
- On-hold
- Done
- Archived

Grouped subtotals:

Group	Included Statuses
WIP	New, To-Do, In Progress, Feedback, Review, Ready to Merge, Final Check, Returned
Stopped	Pending, On-hold
Done	Done, Archived

C. Sprint Execution Efficiency / Original Sprint Composition

Item	Value	%	Explanation
Original Sprint — Carried Over	Count	%	Tickets related to sprint but originally from previous sprint
Original Sprint — Current	Count	%	Tickets originally planned for current sprint
Total Sprint Planned Tickets	Count	100%	Total committed sprint tickets

D. Sprint Completion Summary

Item	Value	%	Explanation
Total Completed Committed Tickets	Count	%	Done/Archived committed tickets by sprint due date
Total Committed Tickets Not Delivered	Count	%	Not Done/Archived committed tickets
Total Committed Tickets	Count	100%	Total committed sprint tickets
Sprint Health Indicator	Status	—	Based on completion and carry-over logic

E. Done Tickets Time Efficiency

Metric	Formula
Cycle Time	Average close/archive timestamp - first In Progress timestamp
Lead Time	Average close/archive timestamp - created timestamp

F. Ticket Status Change Trend

Status	Start of Period	Dates in Between	End of Period	Change	Rate / Week

G. Tracker Change Trend

Tracker	Start of Period	Dates in Between	End of Period	Change	Rate / Week

Tracker rows:

- Bug
- Change Request
- Feature
- Customer Support

H. High-Priority Delivery View

Priority Group	Total High-Priority Tickets	Completed on Time	High-Priority Delivery Rate %	Avg High-Priority Cycle Time	Delayed High-Priority Tickets

I. Daily Agile Board Snapshot

Snapshot Date	Sprint	New	To-Do	In Progress	Feedback	Review	Ready to Merge	Final Check	Returned	Pending / On-hold	Done / Archived	Total

Recommended Visuals

- Progress bar: Sprint Completion Rate
- Stacked bar: WIP / Stopped / Done
- Owner-by-status matrix
- Bar chart: carried-over vs current sprint work
- Trend chart: status movement over period
- High-priority delivery card

Action Triggers

- Sprint completion below expected level
 - High carry-over
 - Large WIP with low Done
 - High Returned count
 - High Pending/On-hold count
 - Many tickets still New/To-Do late in sprint
 - High carried-over share in sprint scope
 - High-priority tickets not completed on time
-

3.3 Team / Ticket Owner Performance Dashboard

Purpose

Provide balanced owner-level performance visibility across commitment, completion, throughput, cycle time, returns, debt, complexity, time utilization, and high-priority responsiveness.

Primary Users

- Project Manager
- Team Lead
- PMO

Main Questions

- Who is carrying the most work?
- Who is completing committed work?
- Who has debt or unfinished commitments?
- Who has high return/rework concentration?
- Who is responsive to high-priority tickets?
- Who is spending time inside vs outside sprint scope?
- Where is owner-level support needed?

Main KPI Cards

KPI	Purpose
Owner Completion Rate (%)	Delivery reliability by owner
Owner Throughput	Tickets completed per week
Owner Debt	Overdue or unfinished work burden
Average Cycle Time	Delivery speed
Average High-Priority Cycle Time	Urgent-work responsiveness
Total Returns	Owner-level return burden
Ticket Idle Time	Activity/update discipline
Sum of Complexity Weight	Workload context

KPI	Purpose
Estimation Accuracy (%)	Planning accuracy by owner
Sprint Logged Time	Owner effort on sprint work
Non-Sprint Logged Time	Owner effort outside sprint scope

Main Tables

A. Ticket Delivery Analysis by Owner

Owner	# Debt	# New Commitment	# Committed Tickets to Finish by End of Period	# Done Committed Tickets	Completion Rate %	# Total Tickets Done During Period	Throughput

B. Done Committed Tickets Analysis by Owner

Owner	# Done Committed Tickets	Sum of Ticket Complexity Weights	R1-R4	Total Returns	Avg Cycle Time for Committed Tickets	Avg Cycle Time for High-Priority Tickets

C. Average Time for Each Status

Status	Average Duration in Status	Average Count of Ticket Became This Status

Status rows:

- To-Do
- In Progress
- Feedback
- Review
- Ready to Merge
- Final Check
- Returned
- On-Hold
- Pending

D. Owner Ticket Idle / No Update Detail

Owner	Active Tickets	Tickets Without Updates	Tickets Without Updates %	Average Idle Time	Highest Idle Ticket

E. Owner Time Allocation

Note: This table is maintained under the Time Utilization & Scope Control section. In the Team / Ticket Owner Performance dashboard, it is used as a secondary cross-view to support owner-level workload and time-allocation analysis.

Owner	Sprint Logged Time	Non-Sprint Logged Time	Total Logged Time	Out-of-Sprint Work Ratio %	Main Non-Sprint Ticket / Activity

Interpretation Rule

Owner performance must be interpreted using multiple signals together. High completion count alone is not enough. The dashboard should consider commitment size, debt, returns, cycle time, high-priority work, complexity context, time allocation, and ticket data quality.

3.4 QA & Returns Dashboard

Purpose

Track internal quality, return behavior, rework effort, first-time pass performance, and return causes.

Primary Users

- QA Lead
- Project Manager
- Team Lead
- PMO

Main Questions

- How many completed tickets passed without return?
- How much work is being returned?
- What are the main return reasons?
- Where is rework time being spent?
- Are returns caused by implementation, QA, review, merge, requirement, or environment issues?

Main KPI Cards

KPI	Purpose
Return Rate (%)	Share of tickets with returns
First-Time Pass Rate (%)	Tickets completed without return
Rework Ratio (%)	Share of effort spent on rework
Total Rework Time	Total rework duration
Tickets with Min 1 Return	Count of completed tickets with return
Tickets with No Return	Count of completed tickets without return
Critical Return Reason Rate (%)	Share of returns caused by serious development / implementation issues

Main Tables

A. Done Tickets Analysis

Item	Count	%
Tickets with Min 1 Return	Count	%
Tickets with No Return	Count	%
Total Done Tickets	Count	100%

B. Returned Tickets Analysis — Count

Return Type	Count	%
R1	Count	%
R2	Count	%
R3	Count	%
R4	Count	%
Total	Count	100%

C. Returned Tickets Analysis — Duration

Rework Type	Duration	%
Development Rework	Duration	%
QA Rework	Duration	%
Review Rework	Duration	%
Merge and Final Check Rework	Duration	%
Total Rework Duration	Duration	100%

D. Work & Rework Ratio

Item	Duration	%
Total Work Duration	Duration	%
Total Rework Duration	Duration	%
Total Durations	Duration	100%

E. Return Reason Analysis

Return Reason	Count	%
Wrong Implementation	Count	%
Incomplete Fix	Count	%
Caused Side Effect	Count	%
Missed Requirement / Misunderstanding	Count	%
Data / Environment Issue	Count	%
Code Quality Issue	Count	%

Merge / Integration Conflict	Count	%
Out-of-Scope Modification	Count	%
Total	Count	100%

Action Triggers

- High return rate
 - Low first-time pass rate
 - High development rework
 - High QA rework
 - Repeated merge/integration conflicts
 - High missed requirement / misunderstanding returns
 - High side-effect / regression return reason
 - Repeated R1–R4 return patterns
-

3.5 Bug Control Dashboard

Purpose

Monitor bug flow, remaining bug pressure, bug source categories, impact rating, and bug concentration by page/module.

Primary Users

- QA Lead
- Project Manager
- Dev Lead
- PMO

Main Questions

- Are bugs being created faster than they are closed?
- What bug categories are increasing?
- What bugs remain at the end of the period?
- Is bug impact increasing or decreasing?
- Which pages/modules have concentrated bug issues?

Main KPI Cards

KPI	Purpose
Bugs Created / Found	New bug pressure
Bugs Resolved / Closed	Bug burn-down output
Remaining Bugs	End-of-period unresolved bug load
Critical Bugs Open	Immediate risk
Bug Leakage (%)	Escaped defect signal
QA Escape Rate (%)	QA detection weakness
Test Coverage Gap Rate (%)	Test scope weakness
Regression Rate (%)	Side-effect / stability weakness
Average Bug Impact Rating	Impact/severity proxy
Start-End Bug Change (%)	Bug backlog movement

Main Tables

A. Bugs Analysis by Category

Bug Category	Bugs at Beginning	%	Bugs Found During Period	%	Bugs Closed During Period	%	Remaining Bugs at End	%	Start-End Change %

Bug categories:

- Design / Requirement Gap
- Side Effect / Regression
- Test Escape
- Test Coverage Gap
- Data / Environment Issue

B. Bug Impact Summary

Item	Beginning	Found	Closed	Remaining
Sum of Bug Impact Rating	Value	Value	Value	Value
Average Bug Impact Rating	Value	Value	Value	Value

C. Related Page Histogram

Page / Module	Bug Count	%	Impact Rating	Risk

D. Critical Bugs Open Detail

Bug ID	Project	Page / Module	Bug Category	Priority / Severity	Impact Rating	Owner	Status	Last Update	Risk

Interpretation Rule

Bug source categories provide diagnostic insight. They should support QA and development improvement actions but should not be mixed with internal return metrics.

3.6 Planning & Estimation Dashboard

Purpose

Evaluate whether sprint and period planning were realistic and whether actual execution aligned with estimates and commitments.

Primary Users

- Project Manager
- Team Lead
- PMO

Main Questions

- Was the sprint overloaded?
- Did actual time exceed estimated time?
- How much work was carried over from previous sprints?
- How many committed tickets were not started?
- Why did tickets carry over?
- Are PM and Owner estimations aligned?
- How much work occurred outside sprint scope?

Main KPI Cards

KPI	Purpose
Planned Tickets	Planned sprint scope
Planned vs Actual Time	Planning accuracy indicator
Estimation Accuracy (%)	Accuracy of estimate vs actual
Commitment Reliability (%)	Delivered / committed
Not Started Count & %	Planned work never started
Carry-over (%)	Planning/delivery burden
Technical Debt Count	Previous unfinished committed work
Missing PM Estimation %	Planning-data weakness
Out-of-Sprint Work Ratio (%)	Scope-control weakness

KPI	Purpose
PM vs Owner Estimation Variance (%)	Planning alignment gap
Owner Estimation Accuracy (%)	Owner estimate vs actual accuracy

Main Tables

A. Planning vs Actual Time Analysis

Item	Value
Sprint Tickets #	Count
Total Estimated Time (h)	Hours
Total Spent Time During Sprint Period (h)	Hours
Deviation (h)	Hours
Deviation (%)	%

B. PM Estimation vs Owner Estimation vs Time Spent

Ticket	Project	Owner	PM Estimation	Owner Estimation	Time Spent	PM vs Owner Variance	Owner Estimate vs Actual Variance	Risk

C. Original Sprint Composition

Item	Count	%	Explanation
Original Sprint — Carried Over	Count	%	Tickets related to sprint but originally from previous sprint
Original Sprint — Current	Count	%	Tickets originally planned for current sprint
Total Sprint Planned Tickets	Count	100%	Total sprint scope

D. Open Planned Tickets Health

Project	Total Open Planned Tickets	Without Owner	Without Owner %	Without Start Date	Without Start Date %	Without Due Date	Without Due Date %	Without PM Estimation	Without PM Estimation %

E. Not Started Tickets

Ticket	Owner	Priority	Sprint	Start Date	Due Date	Status	Estimate	Risk

F. Commitment Reliability Summary

Sprint / Period	Planned Tickets	Committed Tickets	Done Committed Tickets	Commitment Reliability %	Carry-over %	Not Started %

G. Carry-over Reason Analysis

Carry-over Reason	Count	%	Example Tickets	Recommended Action
Planning / Overcommitment	Count	%	Tickets	Improve sprint planning and capacity allocation
PM vs Owner Estimation Gap	Count	%	Tickets	Align PM and owner estimates earlier
Owner Estimate vs Actual Spent Time Gap	Count	%	Tickets	Improve owner estimation or review execution variance
Execution Delay	Count	%	Tickets	Follow up execution blockers and owner progress
External Dependency / Blocker	Count	%	Tickets	Escalate dependency resolution

Action Triggers

- High deviation between estimated and actual time
 - High carried-over share
 - High not-started count
 - Missing PM estimation
 - Missing start date or due date
 - Low commitment reliability
 - Large sprint scope with low completion
 - High out-of-sprint work ratio
 - High PM vs owner estimation variance
-

3.7 Ticket Quality & Data Discipline Dashboard

Purpose

Ensure InfoSystem / Redmine data is reliable enough for planning, execution control, performance analysis, and dashboard interpretation.

Primary Users

- PMO
- Project Manager
- Team Lead

Main Questions

- Can dashboard data be trusted?
- Which projects or tickets have missing required fields?
- Which tickets are active but not updated?
- Are owners, dates, priorities, sprint fields, and estimations properly maintained?

Main KPI Cards

KPI	Purpose
Missing Required Fields (%)	Overall data completeness
Tickets Without Updates (%)	Activity/update discipline
Tickets Missing Owner	Accountability gap
Tickets Missing Start Date	Planning-readiness gap
Tickets Missing Due Date	Deadline-control gap
Tickets Missing PM Estimation	Estimation/planning gap
Tickets Missing Priority	Priority-control gap
Tickets Missing Sprint	Sprint-control gap

Main Tables

A. Open Planned Tickets Health

Date	Project	Total Open Planned Tickets	Without Owner	Without Owner %	Without Start Date	Without Start Date %	Without Due Date	Without Due Date %	Without PM Estimation	Without PM Estimation %
------	---------	----------------------------	---------------	-----------------	--------------------	----------------------	------------------	--------------------	-----------------------	-------------------------

--	--	--	--	--	--	--	--	--	--	--

B. Tickets Without Updates

Ticket	Project	Owner	Status	Priority	Last Update	Days Since Last Update	Sprint	Risk

C. Missing Required Fields Detail

Ticket	Project	Owner Missing	Start Date Missing	Due Date Missing	PM Estimation Missing	Priority Missing	Sprint Missing	Risk

D. Data Reliability by Project

Project	Total Active Tickets	Missing Required Fields %	Tickets Without Updates %	Missing Owner %	Missing Estimation %	Data Reliability Status

Interpretation Rule

If ticket quality is weak, dashboard conclusions should be treated with lower confidence. Weak ticket data should trigger cleanup before strong performance or planning judgments are made.

3.8 Time Utilization & Scope Control Dashboard

Purpose

Track how logged effort is distributed across sprint work, non-sprint work, ticket groups, owners, and reporting periods.

Primary Users

- Project Manager
- Team Lead
- PMO

Main Questions

- How much time was spent on sprint tickets during the selected period?
- How much time was spent outside sprint scope?
- Which owners are spending time outside sprint commitments?
- Which tickets consumed the most time during the period?
- How does actual spent time compare with calendar duration?
- Is logged work aligned with sprint planning?

Main KPI Cards

KPI	Purpose
Sprint Logged Time	Total logged time on sprint tickets
Non-Sprint Logged Time	Logged time outside sprint scope
Out-of-Sprint Work Ratio (%)	Share of effort outside sprint scope
Total Logged Time	Overall effort logged during period
Work Intensity Ratio (%)	Actual spent time compared with calendar duration
Top Non-Sprint Work Owner	Owner with highest outside-sprint effort

Main Tables

A. Spent Time: General vs Sprint Tickets

Date Period	Owner	Sprint Logged Time	Non-Sprint Logged Time	Total Logged Time	Out-of-Sprint Work Ratio %

B. Out-of-Sprint Scope Works

Owner	Ticket	Project	Tracker	Activity	Logged Time	Log Date	Belongs to Sprint?	Sprint	Risk

C. Spent Time on Group of Tickets in Specific Date Period

Ticket	Project	Owner	Sprint	Previous Logged Time	Logged Time During Period	Total Logged Time	Status	Priority

D. Average Spent Time vs Calendar Duration

Ticket	Owner	Start Date	End / Close Date	Calendar Duration	Spent Time	Work Intensity Ratio %	Status	Risk

E. Owner Time Allocation

Owner	Sprint Logged Time	Non-Sprint Logged Time	Total Logged Time	Out-of-Sprint Work Ratio %	Main Activity Type	Risk

F. Activity Distribution in Spent Time Report

Activity	Owner	Sprint Logged Time	Non-Sprint Logged Time	Total Logged Time	% of Total Logged Time	Related Tickets

Recommended Visuals

- Stacked bar: sprint vs non-sprint logged time by owner
- Table: out-of-sprint time logs
- Bar chart: top tickets by logged time
- Scatter chart: calendar duration vs spent time
- KPI card: Out-of-Sprint Work Ratio (%)

Action Triggers

- High out-of-sprint work ratio
- High non-sprint logged time during sprint period
- Low work intensity for committed tickets
- Large time spent on tickets not linked to sprint
- Owner has high logged time but low completed output

4. Alert Rules

The dashboard should include alerts for conditions requiring PM or team action.

Alert	Trigger	Recommended Action
Sprint Completion Risk	Completion below expected progress	PM/team lead review
High Carry-over	Carry-over exceeds threshold	Replanning and root-cause review
Blocked Work	Blocked / Pending / On-hold tickets exist or remain too long	Follow up with owner/team lead
High Return Rate	Return rate exceeds threshold	QA/dev review
High Rework Time	Rework duration exceeds acceptable level	Root-cause analysis
Critical Bug Risk	Critical bugs open	Immediate triage
Bug Growth Risk	Bugs found greater than bugs closed	Bug burn-down action
Not Started Risk	Committed tickets not started	Sprint scope/control review

Planning Deviation	Actual time materially exceeds estimated time	Estimation review
Missing Data Risk	Required fields missing	Data cleanup
No Update Risk	Active tickets not updated	Owner follow-up
High-Priority Delay	High-priority tickets not completed on time	Escalation or priority review
High Owner Debt	Owner has high overdue or unfinished work	Workload review
High Bug Impact	Average bug impact rating exceeds threshold	QA/dev prioritization
High Out-of-Sprint Work	Out-of-sprint work ratio exceeds threshold	Scope control and PM review
Carry-over Reason Concentration	One carry-over reason dominates	Root-cause action by PM/team lead
Low Work Intensity	Spent time is low compared with calendar duration	Investigate waiting, idle time, or blocked work

5. Phase 1 Dashboard Tables Inventory

The following inventory defines the practical dashboard tables required for implementation. These tables are derived from the pilot dashboards and newly required dashboard needs.

ID	Dashboard Page	Table Name	Source	Purpose
T01	PMO Control	Open Technical Tickets Snapshot	Pilot Dashboard	Show open technical workload and distribution by project.
T02	PMO Control	Cross-Project Risk Summary	Master Design	Summarize project/sprint health and risk signals.
T03	PMO Control	Open Tasks / Business Jobs	Pilot Dashboard	Track non-technical open business work.
T04	PMO Control	Open Milestones	Pilot Dashboard	Show milestone progress and risk.
T05	Sprint Delivery	Committed Tickets for Sprint #X	Required Table	Provide retrospective baseline for sprint metrics.
T06	Sprint Delivery	Sprint Tasks Distribution in Statuses	Pilot Dashboard	Show sprint work distribution by status and owner.
T07	Sprint Delivery	Sprint Execution Efficiency / Original Sprint Composition	Pilot Dashboard	Compare current sprint vs carried-over work.
T08	Sprint Delivery	Sprint Completion Summary	Pilot Dashboard	Summarize completion, unfinished work, and health.
T09	Sprint Delivery	Done Tickets Time Efficiency	Pilot Dashboard	Show cycle time and lead time for completed tickets.
T10	Sprint Delivery	Ticket Status Change Trend	Pilot Dashboard	Track status movement over a reporting period.
T11	Sprint Delivery	Tracker Change Trend	Pilot Dashboard	Track tracker-level movement over a reporting period.
T12	Sprint Delivery	High-Priority Delivery View	Master / Required	Track urgent work completion and cycle time.
T13	Owner Performance	Ticket Delivery Analysis by Owner	Pilot Dashboard	Compare owner-level commitment, completion, debt, and throughput.
T14	Owner Performance	Done Committed Tickets Analysis by Owner	Pilot Dashboard	Analyze completed work by owner, complexity, returns, and cycle time.
T15	Owner Performance	Average Time for Each Status	Pilot Dashboard	Identify owner/team bottlenecks by workflow status.
T16	Owner Performance	Owner Ticket Idle / No Update Detail	Master Design	Identify owner-level stale or inactive work.

ID	Dashboard Page	Table Name	Source	Purpose
T17	Owner Performance	Owner Time Allocation	Required Table	Show sprint vs non-sprint logged time by owner. This table is identical to T42 and is reused here as a secondary cross-view.
T18	QA & Returns	Done Tickets Analysis	Pilot Dashboard	Compare returned vs no-return completed tickets.
T19	QA & Returns	Returned Tickets Analysis — Count	Pilot Dashboard	Count returns by return type.
T20	QA & Returns	Returned Tickets Analysis — Duration	Pilot Dashboard	Analyze rework duration by type.
T21	QA & Returns	Work & Rework Ratio	Pilot Dashboard	Compare total work duration and rework duration.
T22	QA & Returns	Return Reason Analysis	Pilot Dashboard / Custom Field	Analyze returns by standardized reason.
T23	Bug Control	Bugs Analysis by Category	Pilot Dashboard	Track bugs by source/category across period states.
T24	Bug Control	Bug Impact Summary	Pilot Dashboard	Summarize bug impact rating across beginning/found/closed/remaining.
T25	Bug Control	Related Page Histogram	Pilot Dashboard	Identify modules/pages with concentrated bugs.
T26	Bug Control	Critical Bugs Open Detail	Master Design	List critical open bugs requiring action.
T27	Planning & Estimation	Planning vs Actual Time Analysis	Pilot Dashboard	Compare estimated time and actual spent time.
T28	Planning & Estimation	PM Estimation vs Owner Estimation vs Time Spent	Required Table	Identify planning and estimation variance.
T29	Planning & Estimation	Original Sprint Composition	Pilot Dashboard	Split current sprint work vs carried-over work.
T30	Planning & Estimation	Open Planned Tickets Health	Pilot Dashboard	Identify missing owner, dates, and estimation on planned work.
T31	Planning & Estimation	Not Started Tickets	Master / Required	List committed tickets never started.
T32	Planning & Estimation	Commitment Reliability Summary	Master Design	Track planned vs done, carry-over, and not-started by sprint.
T33	Planning & Estimation	Carry-over Reason Analysis	Required Table	Identify reasons for moved / unfinished work.
T34	Ticket Quality	Open Planned Tickets Health	Pilot Dashboard	Show planning-field completeness.

ID	Dashboard Page	Table Name	Source	Purpose
T35	Ticket Quality	Tickets Without Updates	Master Design	Identify stale active tickets.
T36	Ticket Quality	Missing Required Fields Detail	Master Design	List tickets missing required planning fields.
T37	Ticket Quality	Data Reliability by Project	Master Design	Summarize data quality by project.
T38	Time Utilization & Scope Control	Spent Time: General vs Sprint Tickets	Required Table	Compare sprint and non-sprint logged effort.
T39	Time Utilization & Scope Control	Out-of-Sprint Scope Works	Required Table	Identify time logs on tickets outside sprint scope.
T40	Time Utilization & Scope Control	Spent Time on Group of Tickets in Specific Date Period	Required Table	Analyze period-specific effort by ticket and owner.
T41	Time Utilization & Scope Control	Average Spent Time vs Calendar Duration	Required Table	Compare actual effort with calendar duration.
T42	Time Utilization & Scope Control	Owner Time Allocation	Required Table	Summarize sprint/non-sprint time by owner.
T43	Time Utilization & Scope Control	Activity Distribution in Spent Time Report	Required Table	Show logged time distribution by activity type.
T44	PMO Control	Feature / Improvement Visibility by Version or Date	Required Table	Track Features and Improvements by version/date, status, owner, and sprint.
T45	Sprint Delivery	Daily Agile Board Snapshot	Required Table	Capture daily sprint board status counts for retrospective sprint-flow analysis.

6. Glossary / Key Definitions

Term	Definition
Metric	A measurable indicator or calculation used to evaluate delivery, quality, planning, execution, bugs, release readiness, or ticket quality.
Dashboard Table	A structured report/table that presents one or more metrics, supporting details, drill-down data, or analytical views.
Dashboard Page / Section	A grouped dashboard area organized around a management or operational purpose.
Main Dashboard Section	One of the eight Target Dashboard sections used as the primary structure for the PMO dashboard.
Supporting / Detail Tab	A secondary view used for drill-down, explanation, analysis, or reuse under one or more main dashboard sections.
Cross-View Table	A table that has one primary dashboard location but is reused by another dashboard section as supporting context.
Target Dashboard Design	The stable designed state of the dashboard system, independent from temporary implementation status in Redmine.
Implementation Tracker	A separate workbook used to compare the Target Dashboard Design with actual Redmine implementation and identify gaps.

7. Implementation Notes

1. Source of Truth

Dashboard data should come from InfoSystem / Redmine.

2. Required Fields

The dashboard depends on reliable owner, project, tracker, sprint, original sprint, status, priority, start date, due date, estimated time, PM estimation, owner estimation, created date, close/archive date, last update, complexity weight, carry-over reason, return reason, bug source, and time-log data.

3. Workflow History Requirement

Cycle time, lead time, blocked duration, return count, rework duration, average time in status, and actual start date require reliable status transition history.

4. Time-Log Requirement

Planned vs actual time, estimation accuracy, rework ratio, sprint logged time, non-sprint logged time, out-of-sprint work ratio, and work intensity require consistent time logging.

5. Sprint Baseline Requirement

Committed Tickets for Sprint #X must be retrospectively available. Original Sprint should be preserved and should not be overwritten when tickets move to a future sprint.

6. Return Tracking Requirement

Return Rate, First-Time Pass Rate, Return Reason Analysis, R1–R4 counts, and Rework Time require reliable return-loop tracking.

7. Bug Classification Requirement

Bug source/category, impact rating, critical bugs, leakage, regression, test escape, and related page histogram require consistent bug classification.

8. Carry-over Reason Requirement

Carry-over Reason should use controlled values, not free text, so that planning and execution root-cause reports remain reliable.

9. Blocked vs Stopped Logic

Blocked Tickets Count should use the official blocked status if available. Pending and On-hold should be shown as stopped-work indicators. If the workflow treats Pending/On-hold as blocked states, this mapping must be explicitly confirmed.

10. Technical Debt vs Owner Debt

Technical Debt Count means unfinished committed tickets from previous sprints. Owner Debt means overdue or unfinished work assigned to a specific owner. These are related but should not be merged unless definitions are intentionally aligned.

11. High-Priority Logic

High-priority metrics should use approved priority values, such as Immediate, Hotfix, Urgent, or other officially defined high-priority labels.

12. Critical Bug Logic

Critical Bugs Open should be based on official severity/priority. If formal severity is not available, Bug Impact Rating may be used as a support indicator.

13. Dashboard Validation Period

After implementation, the dashboard should be validated for 2–3 weeks against real ticket behavior to confirm formulas, filters, workflow usage, and data reliability.

14. Phase 1 Scope Rule

The dashboard may include supporting diagnostic tables when data is available, but the top-level control layer should remain simple, actionable, and easy to understand.

15. Performance Interpretation Rule

Ticket-owner performance views should support improvement, coaching, follow-up, and workload balancing. They should not be used as automatic performance judgment without PM review and data-quality validation.

16. Implementation Alignment Workbook

The Word document and Metrics Excel define the stable Target Dashboard Design. Actual Redmine implementation, current dashboard pages, supporting tabs, implementation status, and remaining gaps are tracked separately in the implementation workbook:

PMO_Dashboard_Implementation_V14HA.xlsx

8. Final Expected Outcome

At the end of Phase 1 implementation, the PMO will have:

- real-time visibility into delivery, execution, quality, bugs, planning, ticket quality, and time utilization
 - a stable dashboard structure based on stakeholder needs
 - a reliable metrics table aligned with dashboard tables
 - traceability between pilot dashboards, required tables, and Redmine implementation
 - consistent drill-down from KPI cards to ticket-level evidence
 - a clear implementation alignment reference for Redmine dashboards
 - a baseline for Phase 2 diagnostic expansion
-

The End